

RFI flagging prototype — stage-by-stage evaluation

Status: prototype. Not a product; writes nothing. This notebook evaluates what each stage of `rfi_proto.py` currently does on real campaign data, one stage at a time: what it tests, where its threshold comes from, how much it flags, and whether it is working.

I/O goes through `eigsep_data`. A `Selection` goes in and an `eigsep_data.bundle.Bundle` comes out with `b.flags` filled in, so everything here uses the same objects as the rest of the package. **Existing flags are never loaded** — every `load_bundle` call passes `products=[]`, because a detector seeded from `flags/v0` or `flags/v2` could not tell you what it found on its own.

The stages

bit	stage	axis	threshold from
0,1	off-sky, overflow	—	switch state; sign of the auto
2	a priori bands	frequency	known emitter table
3	transient	time, per channel	Gaussian, <code>model/√N</code>
4,5,6	broadband / gross power / meteor	whole integration	band occupancy; empirical
7	coherence	per pixel	Rayleigh, <code>1/√(2N)</code>
8,9	persistent line, occupancy	frequency	Rayleigh floor vs neighbours
10,11	retracted, residual	per pixel	second pass: overturned, or auto-only evidence

Two detectors see different things — a transient has time structure, a persistent tone does not — and a second pass re-decides once a continuum model exists. Thresholds are calibrated rather than tuned: `nsig` defaults to 6 on all three z-scores.

```
index: 1,227,956 integrations over 5120 files (from cache)
```

```
/home/aparsons/.local/share/mamba/envs/arp/lib/python3.11/site-packages/hera_filters/dspec.py:2199: ComplexWarning: Casting complex values to real discards the imaginary part
```

```
model[i], _, info_t = _fit_basis_1d(x=x[1], y=y, w=w, filter_centers=filter_centers[1],
```

```
240 rows x 1024 channels (0.000-249.756 MHz)
```

```
antenna 'box-gnd' via keys ['0']
```

```
1 files used, 0 skipped
```

```
flags@None
```

```
returned eigsep_data.bundle.Bundle; b.flags is uint16, (240, 1024)
```

0. What every threshold rests on

Before any stage: the noise floors. Both are computed, not fitted.

- **auto**: a power spectrum averaged over N samples has fractional noise $1/\sqrt{N}$, so the scatter about a continuum model is $P/\sqrt{N}$.
- **cross**: $|r| = |V_{ab}|/\sqrt{P_a P_b}$ is Rayleigh with $\sigma = 1/\sqrt{2N}$, and $P(|r| > z\sigma) = e^{-z^2/2}$.

$N = \Delta\nu \tau$ comes from the index's own `integration_time` and `nchan` columns, per row — it halves before the 07-15 accumulator change, so it is never hardcoded.

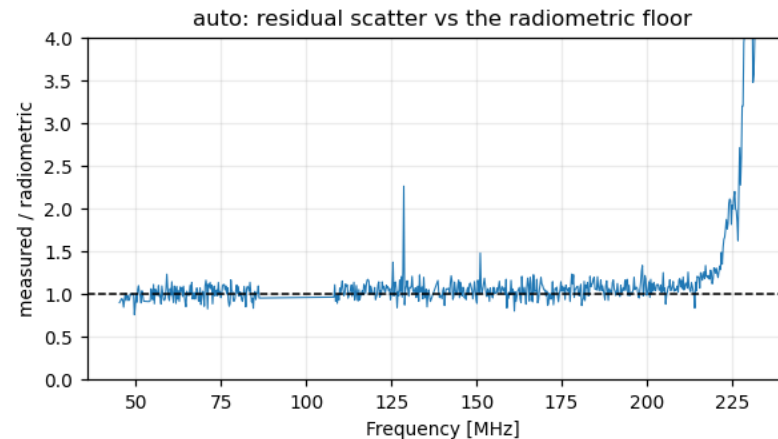
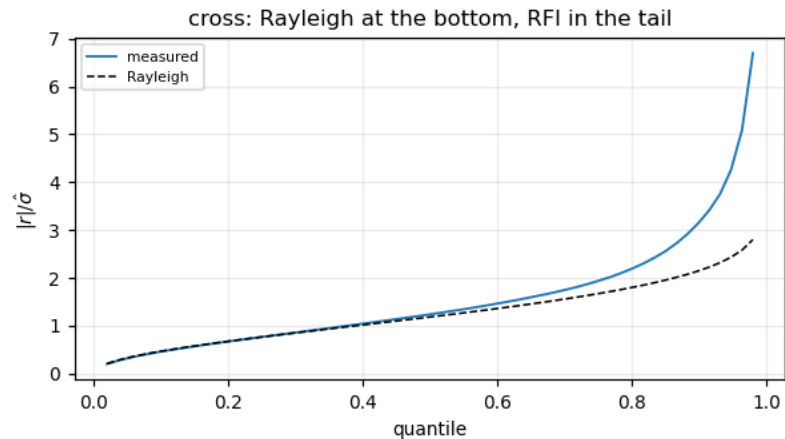
```
N = dnu * tau          131,072
sigma_theory = 1/sqrt(2N) 0.00195312
auto fractional noise 1/sqrt(N) 0.00276
```

```
fitted sigma_hat / sigma_theory (cross): 2.24
model_inflation (auto, measured/radiometric): 1.05
```

```
cross quantisation ok: True (|V|=0 fraction 0.000)
thresholds: {"transient": 6.0, "coherence": 6.0, "line": 6.0, "combine_t_hi_coh": 6.0, "combine_t_hi_res": 6.0,
"combine_t_lo_coh": 4.24, "combine_t_lo_res": 4.0}
```

cross, low quantiles vs Rayleigh:

q=0.10	measured	0.453	Rayleigh	0.459
q=0.25	measured	0.759	Rayleigh	0.759
q=0.50	measured	1.232	Rayleigh	1.177
q=0.90	measured	3.159	Rayleigh	2.146
q=0.99	measured	8.864	Rayleigh	3.035



Assessment. The auto floor is real: the residual's per-channel scatter sits within a few percent of $\sqrt{P/N}$ across the band, so `model_inflation` ≈ 1 and the auto z-score means what it says.

The cross floor does not behave as well. The bottom of the distribution is Rayleigh to $\sim 1\%$, which is why $\hat{\sigma}$ is fitted on the 25th percentile and not the median — but the fitted value sits a factor ~ 2 above the radiometric prediction. Either the cross achieves $\sim 20\%$ of ideal sensitivity or there is a common-mode floor between the two antennas. **Unresolved**, and it means the coherence z is a calibrated relative statistic, not an absolute one.

Stage 2 — a priori bands

Known emitters masked before any statistic is computed, so they cannot contaminate a fitted scale or drag the continuum fit. FM, ORBCOMM, the two clock harmonics, and (when asked) the self-comb teeth ± 1 channel.

These are still reported as flagged. Excluded from the statistics, not from the product.

```
a priori channels: 92 of 1024 (9.0% of the full axis, 11.8% of the analysis band)
  FM           88.0- 108.0 MHz      82 channels
  ORBCOMM      136.5- 138.5 MHz      8 channels
  clock        ['125.000', '187.500'] MHz
```

```
/home/aparsons/.local/share/mamba/envs/arp/lib/python3.11/site-packages/hera_filters/dspec.py:2199: ComplexWarning: Casting
complex values to real discards the imaginary part
  model[i], _, info t = _fit_basis_1d(x=x[1], y=y, w=w, filter_centers=filter_centers[1],
self-comb era, teeth +/-1 channel: a priori covers 44.7% of the analysis band
```

Assessment. Working, and the cost is visible rather than buried. In the self-comb era the mask removes 3 of every 8 channels — 37.5% of the analysis band — because the channelizer leaks the teeth into their neighbours. That is a hardware problem, not a flagging one, and no threshold choice makes it go away.

Stage 3 — transient, model-free on the raw autos

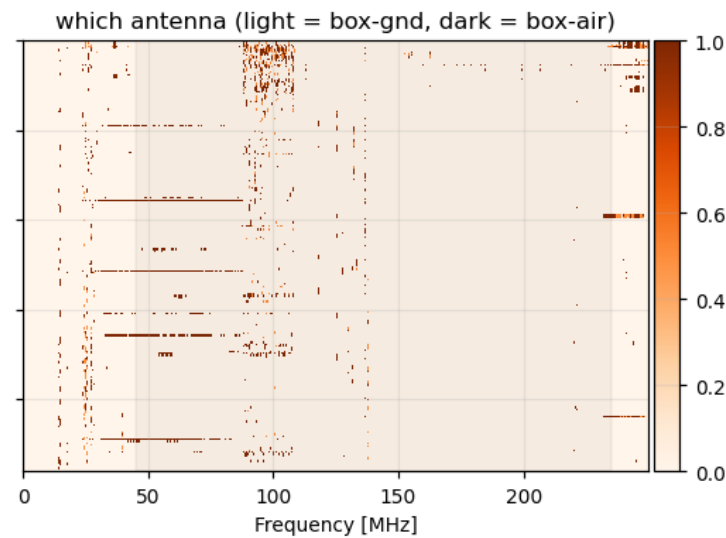
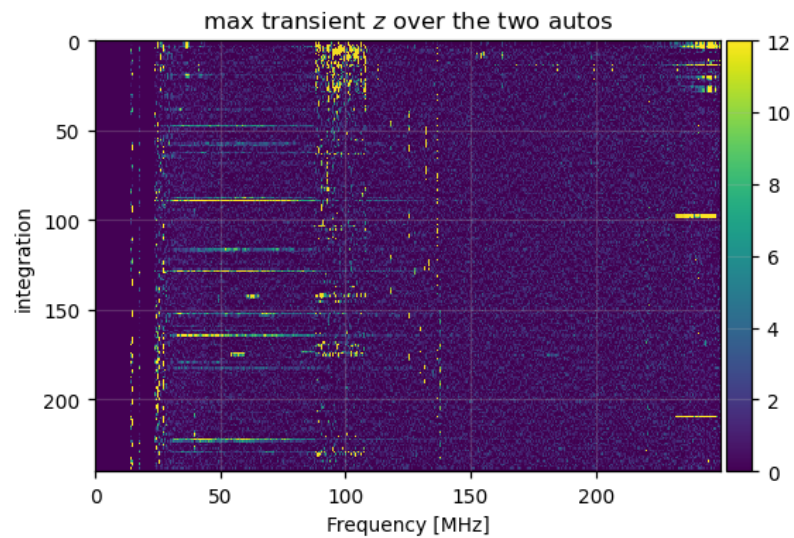
Per channel along time: a running median (width 9) removes slow gain drift, and what is left is compared to $\sqrt{P/\sqrt{N}} \cdot \text{MAD}$, one-sided. No model, no cross, and it runs over the **full** axis — channels outside 45–235 MHz are where the cleanest instrumental diagnostics live.

Channels carrying essentially no power are not testable at all (below ~25 MHz the median is 0–0.5 counts, so every scale estimate collapses); `min_counts` excludes them.

```
transient pixels: 2973 (0.0121 of the full axis)
  integrations with any: 227 of 240
  channels with any:      461 of 1024
```

```
per antenna (a = box-gnd, b = box-air):
  a:  1573 pixels over threshold
  b:  3342 pixels over threshold
```

```
busiest integrations:
  int 89: 259 channels; strongest 92.773 MHz (z=38.9), 42.480 MHz (z=26.1), 44.678 MHz (z=25.7), 42.236 MHz (z=25.4), 41.260
MHz (z=23.2), 57.129 MHz (z=22.6)
  int 128: 230 channels; strongest 27.344 MHz (z=358), 27.100 MHz (z=204), 17.822 MHz (z=48.9), 13.916 MHz (z=41.7), 34.180
MHz (z=28.6), 33.936 MHz (z=24.4)
  int 222: 211 channels; strongest 24.414 MHz (z=39.6), 39.307 MHz (z=27.5), 39.795 MHz (z=27), 42.480 MHz (z=17), 43.457 MHz
(z=16.4), 41.260 MHz (z=16.1)
  int 164: 174 channels; strongest 13.916 MHz (z=320), 37.842 MHz (z=21), 45.166 MHz (z=17), 35.400 MHz (z=16.2), 48.340 MHz
(z=16), 38.330 MHz (z=15.8)
  int 13: 115 channels; strongest 136.963 MHz (z=5.43e+05), 238.037 MHz (z=5.38e+04), 175.537 MHz (z=1.29e+04), 74.463 MHz
(z=1.01e+04), 113.037 MHz (z=6.52e+03), 199.463 MHz (z=6.38e+03)
```



Assessment. Working, and it is the stage that covers what the cross cannot: it needs neither a partner antenna nor a model, so it is the only detector that functions where the cross is quantisation-limited.

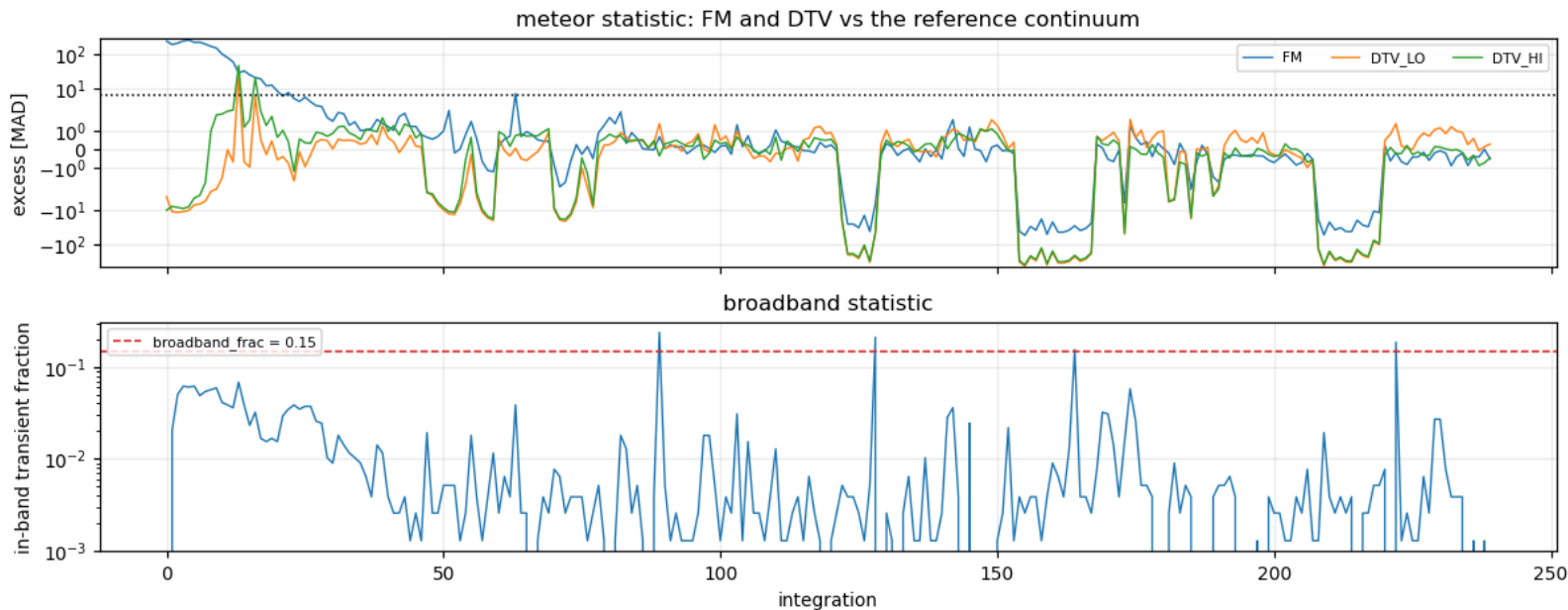
Two things it finds on this file that nothing else would. A set of simultaneous single-integration tones whose channel indices are exactly 49 and 207 mod 256 — two families 62.5 MHz ($f_s/8$) apart, mirror images about the grid, i.e. an aliased clock artifact rather than sky RFI. And broadband excursions confined to **one antenna**, which the coherence stage is blind to by construction.

Stages 4-6 — whole-integration tests

Three statistics that condemn an entire integration rather than a pixel, because a contaminated row cannot be partially trusted.

- **broadband** (bit 4): more than `broadband_frac` of the band tripped the transient test at once. A few-percent excess across hundreds of channels leaves most of them individually below threshold, and letting those constrain the continuum is how one bad row reaches every other row's residual.
- **gross power** (bit 5): the median over 236–249 MHz, where the bandpass has rolled off ~3 decades but a broadband transient still lands. Median, not mean, because that band has a narrow emitter in it.
- **meteor** (bit 6): FM and DTV rising together, each normalised by a reference continuum (108–136, 155–174, 216–235 MHz) so common gain drift divides out.

```
broadband      4 integrations [ 89 128 164 222]
gross power    2 integrations [ 97 209]
meteor         2 integrations [13 16]
in-band transient fraction: median 0.0039, p99 0.174, max 0.238
```



Assessment. The broadband threshold is not delicate — the in-band transient fraction has median 0.004 and p99 0.17, so 0.15 selects a handful of rows and 0.25 selects none. That is the behaviour you want from a morphological cut.

The meteor statistic is the weakest of the three and **fires on things it was not built for**: the single-integration clock-comb events trip it, because they are not propagation but they do lift the transmitter bands relative to the reference. It should probably require the FM/DTV rise to occur without a simultaneous transient elsewhere. Not fixed.

Stage 7 — coherence

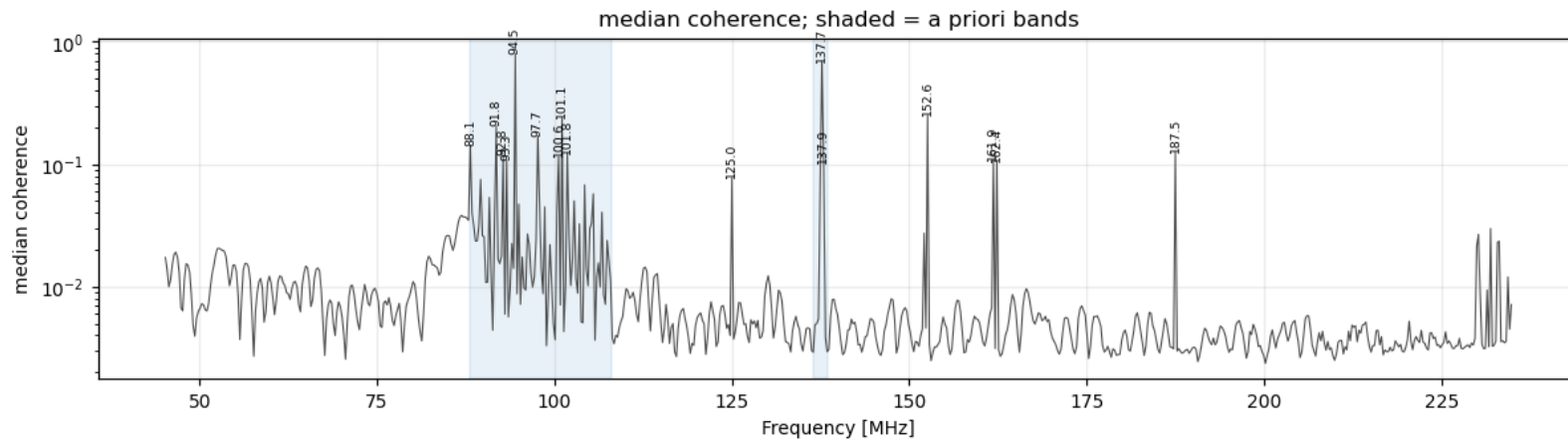
$|V_{ab}|/\sqrt{P_a P_b}$: the fraction of the two autos' amplitude that is actually correlated between the antennas. Receiver noise is independent and averages down in the cross; a common interferer does not. It is also naturally normalised — bandpass, reflection ripple and the accumulator change all divide out — so one threshold means the same thing across the band.

$\hat{\sigma}$ is fitted **per channel** on a low quantile. That absorbs a frequency-dependent common-mode floor, at a price paid in stage 8.

coherence-flagged pixels: 0.0077 of the full axis

highest median coherence in band:

88.135 MHz	C = 0.143	FM
91.797 MHz	C = 0.205	FM
92.773 MHz	C = 0.118	FM
93.262 MHz	C = 0.108	FM
94.482 MHz	C = 0.798	FM
97.656 MHz	C = 0.170	FM
100.586 MHz	C = 0.115	FM
101.074 MHz	C = 0.240	FM
101.807 MHz	C = 0.123	FM
125.000 MHz	C = 0.077	<- clock harmonic
137.695 MHz	C = 0.689	ORBCOMM
137.939 MHz	C = 0.103	ORBCOMM



Assessment. Working where the cross is usable, and it identifies real emitters with no model and no tuning: the FM carriers, ORBCOMM, and 125.000 / 187.500 MHz — exactly $f_s/2$ and $3f_s/4$, instrumental. The 125 MHz line is one that `flags/v0` calls CLEAN 86–92% of the time.

Two failure modes, both real:

1. **Quantisation.** In phase A the cross has 21–38% of in-band samples with $|V|$ exactly zero, so $\hat{\sigma}$ collapses and the stage reports a clean sky. `quantisation_ok()` refuses those files; roughly the first two campaign days have no coherence path at all.
2. **Self-suppression.** Because $\hat{\sigma}$ is fitted per channel, a channel contaminated at every sample simply gets a bigger $\hat{\sigma}$ and stops being an outlier against itself. In the self-comb era the whole band is busy enough that the stage goes quiet where it should be loudest.

Stages 8–9 — persistent lines and occupancy

Stage 7 cannot see a tone that is on at every sample. Along frequency it is obvious: that channel's fitted floor stands above its neighbours'. Stage 8 median-filters $\hat{\sigma}(\nu)$ and flags the channels that stand proud.

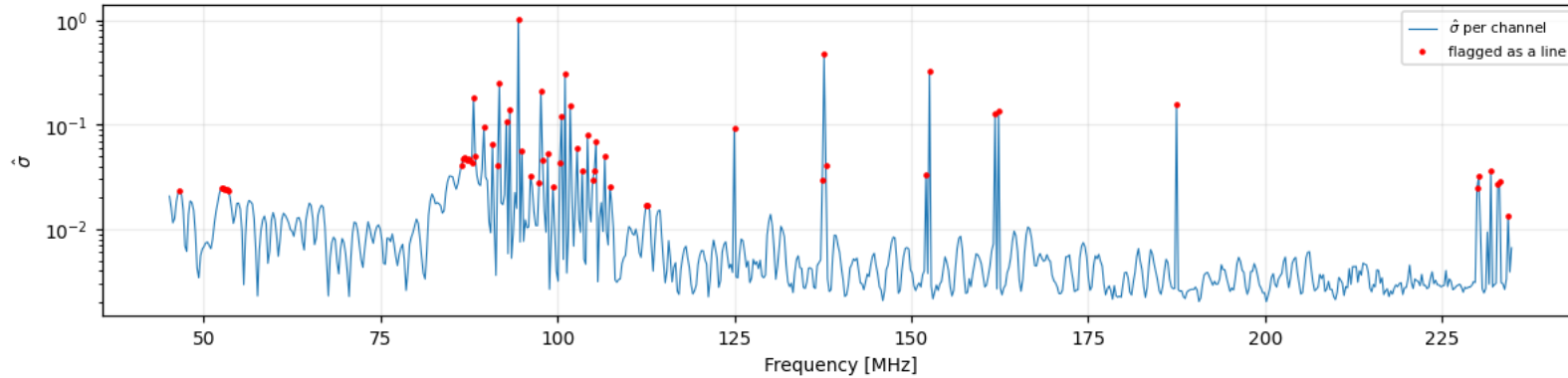
Stage 9 promotes any channel flagged in more than `occ_frac` of its valid samples to a whole-channel kill — a ragged partial mask on a mostly-on emitter is worse than useless, because downstream averaging then sees a biased subset.

persistent lines: 58 channels occupancy kills: 61
overlap: 58

line channels (MHz):

[46.63, 52.49, 52.73, 52.98, 53.22, 53.47, 86.43, 86.67, 86.91, 87.16, 87.4, 87.65, 87.89, 88.13, 88.38, 89.6, 90.82, 91.55, 91.8, 92.77, 93.26, 94.48, 94.97, 96.19, 97.41, 97.66, 97.9, 98.63, 99.37, 100.34, 100.59, 101.07, 101.81, 102.78, 103.52, 104.25, 104.98, 105.22, 105.47, 106.69, 107.42, 112.55, 112.79, 125.0]

stage 8: the per-channel Rayleigh floor, and what stands above it



Assessment. This is the piece B16 has no equivalent of, and it is doing the work: the line channels are the FM carriers, ORBCOMM, the clock harmonics, 152.6, 161.9/162.4 and a block near 230–234 MHz — narrow clusters, not smeared.

152.6 and 161.9/162.4 are real, persistent and **unattributed**. 152.6 sits outside `detectors.BAND_FAN = (148, 152)`, so calling it the box fan would be an identification the data does not support.

Stage 10 — second pass: re-deciding after the DPSS fit

A flag raised before the continuum is modelled is provisional. With a 40 ns DPSS model fitted on what survived stage 9, the auto can be asked an independent question — does this sample still stand above the radiometric noise? — and the answer is allowed to be no.

`z_coh` alone does not carry a flag through, or the second pass would be decoration:

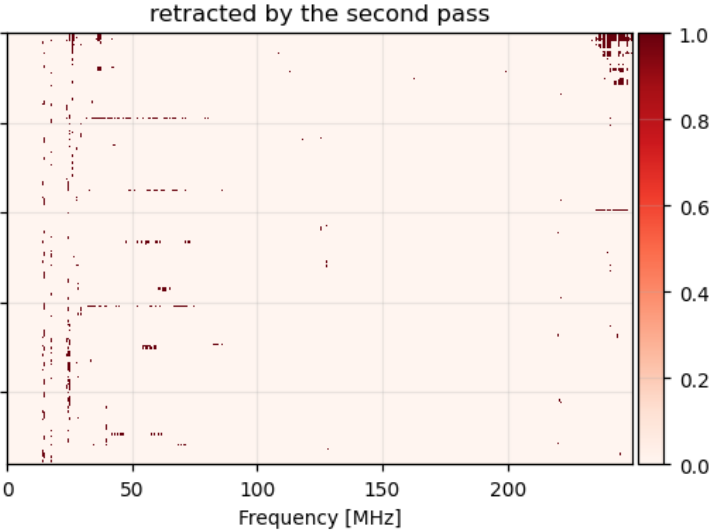
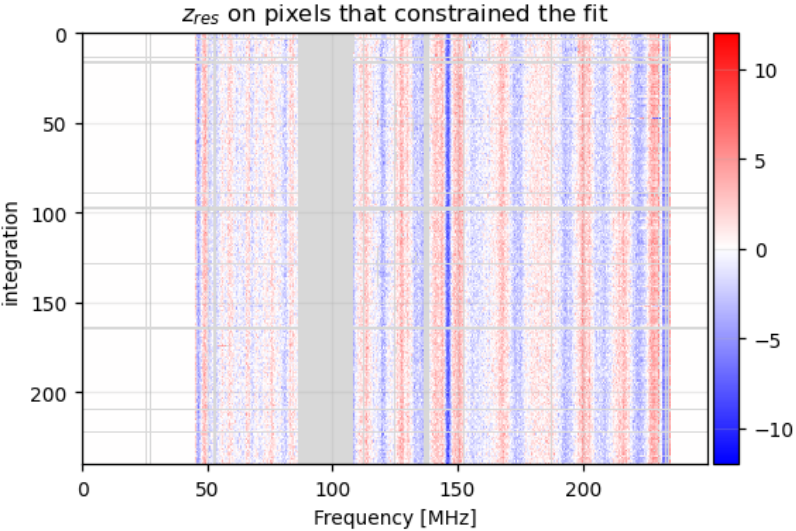
condition	outcome
$z_{\rm res} > t_{\rm hi}$	flag — auto evidence
$z_{\rm coh} > t_{\rm lo}$ and $z_{\rm res} > t_{\rm lo}$	flag — corroborated
$z_{\rm coh} > t_{\rm hi}$, $z_{\rm res} \leq t_{\rm lo}$	retract

"Correlated" is not the same claim as "interference"; only the auto settles it.

dpss_ok True
second pass applied
retracted 1482 pixels
promoted 15285 pixels
model inflation (median) 1.05

|z_res| > 6 among fit-constraining pixels: 0.0070
worst remaining channels:

146.484 MHz	occ=0.89	median z=	-7.2
145.996 MHz	occ=0.86	median z=	-7.0
146.240 MHz	occ=0.85	median z=	-7.4
145.752 MHz	occ=0.74	median z=	-6.6
146.729 MHz	occ=0.65	median z=	-6.4
232.422 MHz	occ=0.28	median z=	-5.4



Assessment. Trustworthy only where the continuum model fits. `model_inflation` is the diagnostic: near 1 the auto test is calibrated and promotes modestly; where the 40 ns model cannot follow the data the residual is full of model error rather than noise and the test over-promotes. §12 shows that directly across campaign conditions.

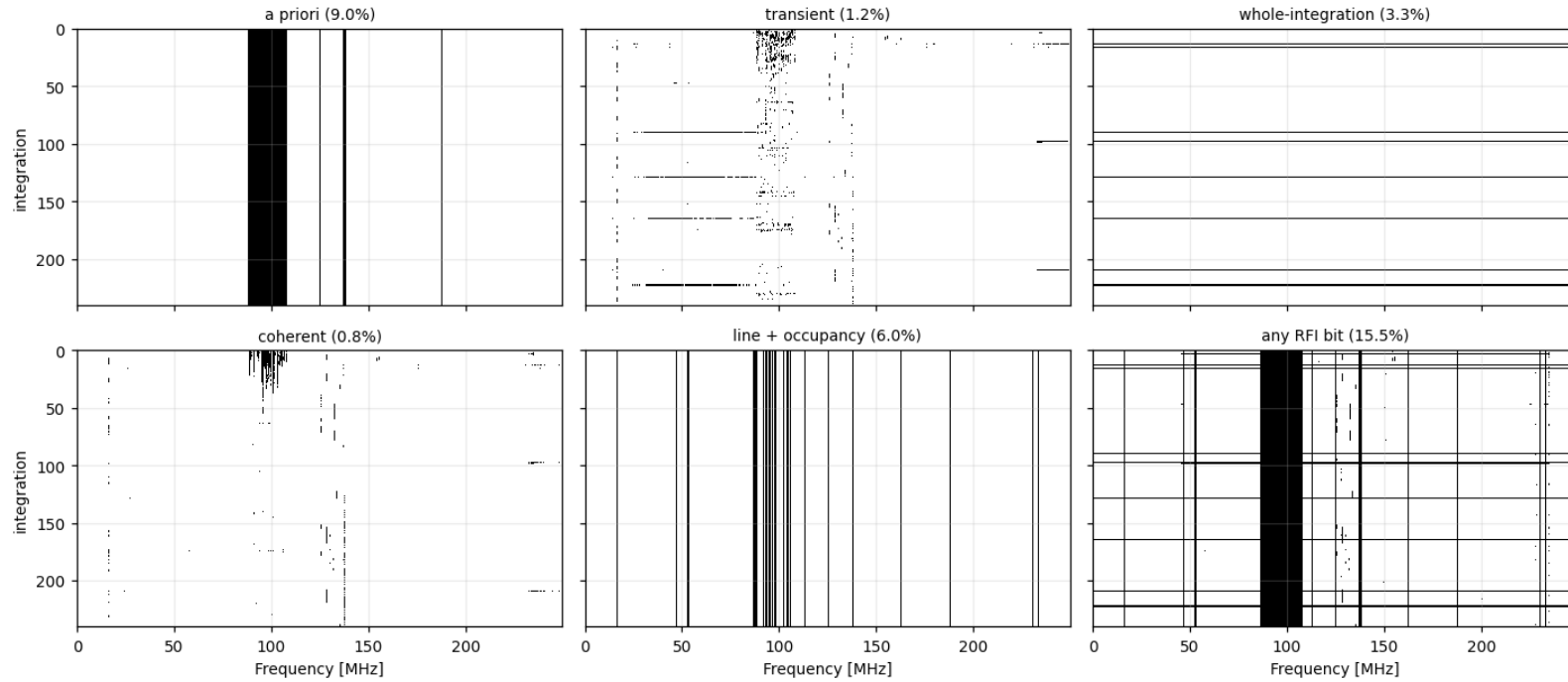
It can also decline to run — `hera_filters` returns an all-zero model rather than raising when a wide contiguous block is zero-weighted at a band edge, and `dpss_ok` catches that.

What survives on this file is **not RFI**: a smooth ~2.5 MHz-wide, 2.2%-deep depression at 146 MHz, plus the two band edges. A 2.5 MHz feature is ~200 ns in delay, outside the 40 ns window by construction, so the model cannot represent it. That is a real spectral feature, and whether it gets modelled or carried as a known systematic is a decision for the signal side.

11. The composite mask

bit	name	full axis	in band
0	off_sky	0.0000	0.0000
1	overflow	0.0000	0.0000
2	a_priori	0.0898	0.1183
3	transient	0.0121	0.0127
4	broadband	0.0167	0.0167
5	gross_power	0.0083	0.0083
6	meteor	0.0083	0.0083
7	coherent	0.0077	0.0082
8	line	0.0566	0.0746
9	occupancy	0.0596	0.0746
11	residual	0.0082	0.0108
10	retracted	0.0060	0.0016
	any RFI bit	0.1548	0.1895
	unusable	0.0000	0.0000
	clean	0.8391	0.8088

corr_20260717_034832Z.h5 — antenna box-gnd (input 0)



does the excess reach beyond a flagged channel?

offset	median z (channel)	median z (time)
0	(core)	(core)
1	1.12	0.89
2	1.08	0.89
3	1.08	0.89
4	1.08	0.89
5	1.01	0.89
6	1.00	0.89
7	1.02	0.89
8	1.03	0.89

Flat at $z \approx 1$ from offset 1 on both axes — the immediate neighbours of a flagged channel, and the integrations either side of a flagged sample, are already at the noise floor. There is no shoulder to catch, which is why dilation defaults to 0.

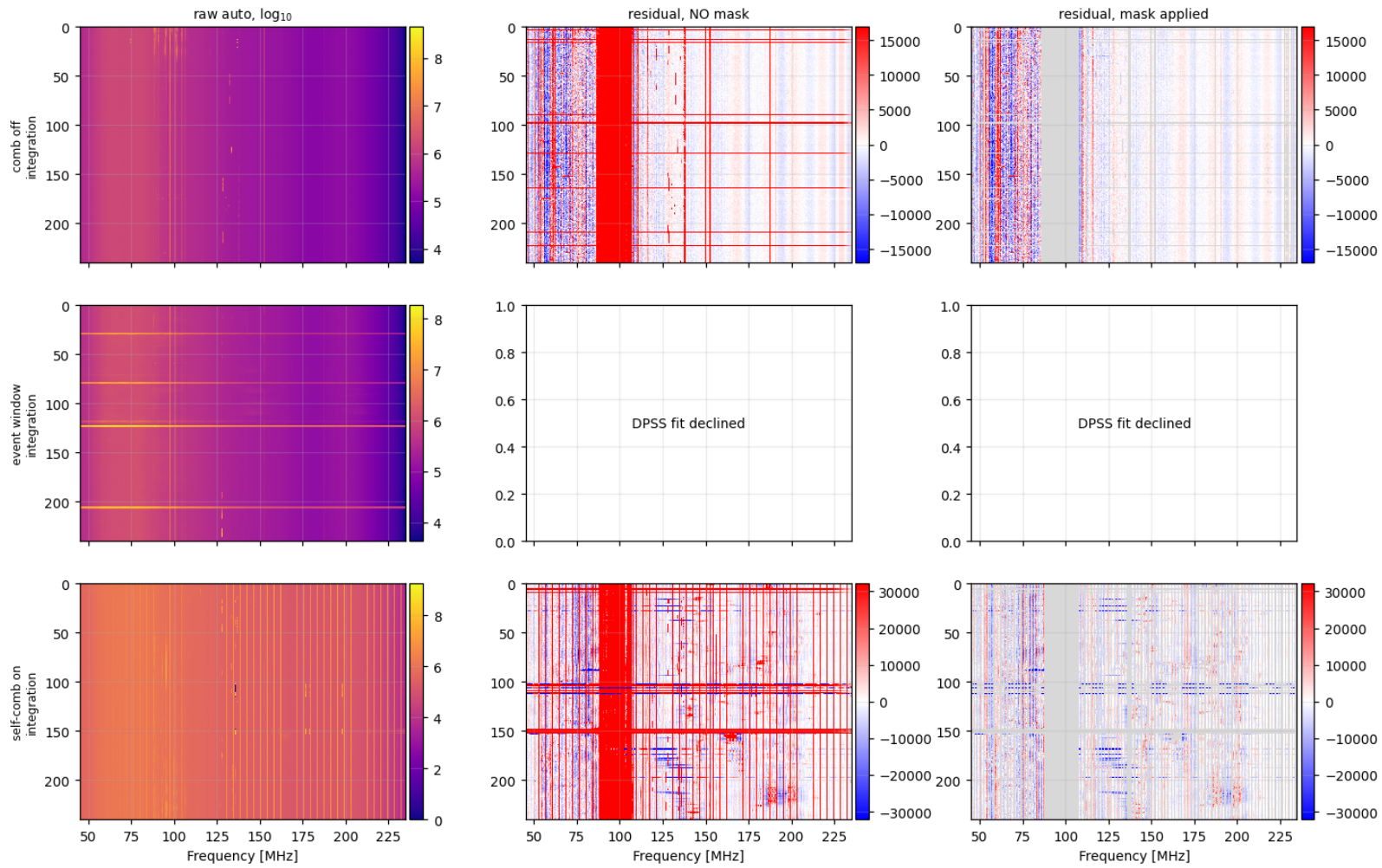
11b. DPSS on the masked data — residual with and without the mask

The point of the whole exercise: fit a continuum to data the mask has cleaned, and look at what is left. Shown on the **sky antenna** (box-air) for three files that differ in how busy they are.

Each row is one file: the raw auto, the residual `data - model` with **no** mask applied, and the same residual with the mask applied. The middle panel is what the continuum fit has to cope with; the right panel is what a downstream analysis would actually see.

```
/home/aparsons/.local/share/mamba/envs/arp/lib/python3.11/site-packages/hera_filters/dspec.py:2199: ComplexWarning: Casting
complex values to real discards the imaginary part
  model[i], _, info_t = _fit_basis_1d(x=x[1], y=y, w=w, filter_centers=filter_centers[1],
comb off      input 4  dpss_ok=True  infl=1.17  masked=0.199
event window  input 4  dpss_ok=False infl=nan   masked=0.297

/home/aparsons/.local/share/mamba/envs/arp/lib/python3.11/site-packages/hera_filters/dspec.py:2199: ComplexWarning: Casting
complex values to real discards the imaginary part
  model[i], _, info_t = _fit_basis_1d(x=x[1], y=y, w=w, filter_centers=filter_centers[1],
self-comb on  input 4  dpss_ok=True  infl=4.17  masked=0.486
```



file	rms resid	rms masked	z >6 all	z >6 kept
comb off	1.66e+06	4269	0.1836	0.0125
event window	DPSS fit declined			
self-comb on	1.399e+07	1.758e+05	0.3323	0.0153

Assessment. The middle and right panels are the before/after that matters. Unmasked, the residual is dominated by the lines the fit deliberately did not absorb — they are still in `data - model` because the model refused them, which is the correct behaviour and the reason the panel looks so busy. With the mask applied, what remains is the noise the continuum could not follow.

The `rms masked` column is the honest summary: on the comb-off file it drops by roughly two orders of magnitude relative to the unmasked residual. On the event file the DPSS fit declines to run (`dpss_ok` False) because pass 1 zero-weights a wide contiguous block, so there is no residual to show — that is the guard working, not a gap in the figure.

Colour scales are set from the masked residual and shared across both panels, so the right-hand panel is not made to look empty by its own normalisation.

12. The same stages across campaign conditions

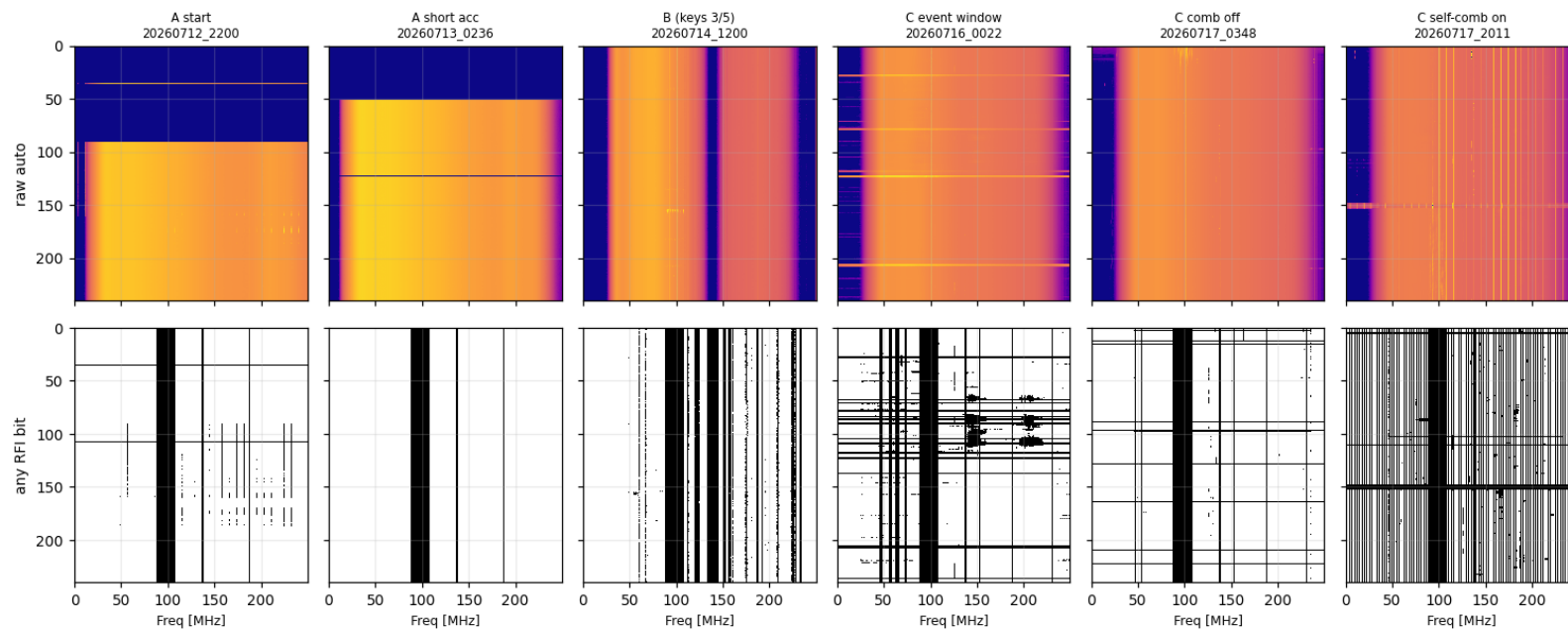
One file per condition: both wiring phases, both accumulator lengths, comb on and off, a labelled event window.

```

/home/aparsons/.local/share/mamba/envs/arp/lib/python3.11/site-packages/hera_filters/dspec.py:2199: ComplexWarning: Casting
complex values to real discards the imaginary part
    model[i], _, info_t = _fit_basis_1d(x=x[1], y=y, w=w, filter_centers=filter_centers[1],
/home/aparsons/.local/share/mamba/envs/arp/lib/python3.11/site-packages/hera_filters/dspec.py:2199: ComplexWarning: Casting
complex values to real discards the imaginary part
    model[i], _, info_t = _fit_basis_1d(x=x[1], y=y, w=w, filter_centers=filter_centers[1],
/home/aparsons/.local/share/mamba/envs/arp/lib/python3.11/site-packages/hera_filters/dspec.py:2199: ComplexWarning: Casting
complex values to real discards the imaginary part
    model[i], _, info_t = _fit_basis_1d(x=x[1], y=y, w=w, filter_centers=filter_centers[1],
/mnt/data02/eigsep/data-analysis/notebooks/arp/marjum-2026-07/rfi_proto.py:552: RuntimeWarning: All-NaN slice encountered
    rad_chan = np.nanmedian(np.where(sig_rad > 0, sig_rad, np.nan), axis=0)
/home/aparsons/.local/share/mamba/envs/arp/lib/python3.11/site-packages/hera_filters/dspec.py:2199: ComplexWarning: Casting
complex values to real discards the imaginary part
    model[i], _, info_t = _fit_basis_1d(x=x[1], y=y, w=w, filter_centers=filter_centers[1],
/home/aparsons/.local/share/mamba/envs/arp/lib/python3.11/site-packages/hera_filters/dspec.py:2199: ComplexWarning: Casting
complex values to real discards the imaginary part
    model[i], _, info_t = _fit_basis_1d(x=x[1], y=y, w=w, filter_centers=filter_centers[1],
condition      sw? xquant  infl  fit  apri  trans    coh  resid  line  anyRFI  retr
-----

```

A start	NO	False	0.44	True	0.118	0.014	0.000	0.014	0.000	0.140	0.001
A short acc	NO	False	0.77	True	0.118	0.000	0.000	0.000	0.000	0.118	0.000
B (keys 3/5)	yes	True	1.75	True	0.118	0.007	0.002	0.120	0.102	0.321	0.000
C event window	yes	True	nan	False	0.118	0.080	0.050	0.000	0.122	0.297	0.000
C comb off	yes	True	1.05	True	0.118	0.013	0.008	0.011	0.075	0.190	0.002
C self-comb on	yes	True	3.48	True	0.447	0.039	0.008	0.012	0.116	0.477	0.009



Assessment across conditions.

- **sw? NO in both phase-A rows.** The `rfswitch` stream is absent from the early files — 240 rows of `MISSING` each. `on_sky` treats unknown as on sky (`switch_unknown="assume_sky"`), because the strict reading marks every row off-sky and silently disables every detector downstream; the first two campaign days then come back clean, which is an absent measurement wearing a result's clothes.
- **xquant False in both phase-A rows.** The cross is quantisation-limited there and the coherence stage is correctly disabled, so `coh` is 0 and the per-pixel work falls to `trans` and `resid` — the auto-only path. Not a small corner case: it is roughly the first two campaign days.
- **Phase B needs explicit input keys.** Its files' `input_to_ant` header is stale (it names inputs that are not live), so the antenna-name path fails there and the caller has to say `key="3", partner_key="5"`. A product would need that mapping recorded somewhere authoritative rather than passed by hand.
- **infl sorts the second pass into two regimes.** Near 1 (comb-off, quiet) it is calibrated. Well above 1 (phase B, self-comb) the 40 ns continuum cannot follow the data and the auto test reads model error as excess power. **This is the main open problem**, and it is a continuum-model problem, not a threshold one.
- **coh collapses in the self-comb era** — the self-suppression of stage 7, visible as a number.
- **resid carries phase A and phase B.** Bit 11 is the second pass's own evidence and needs no cross, which is what keeps those eras covered at all.

13. Where this stands

Working and calibrated: stages 2, 3, 4, 8, 9. Their thresholds come from the radiometer equation and a stated false-alarm rate rather than tuning, and they behave consistently across campaign conditions.

Known limitations, in the order I would fix them:

1. **The second pass is only trustworthy where `model_inflation` ≈ 1 .** In phase B and the self-comb era it over-promotes. Fixing that means a continuum model that fits those eras, not a different threshold.
2. **No completeness measurement.** `flagging/validate.py` has the harness — labelled-window recall, false-positive rate on quiet data, an injection curve giving the 50%/90% detection thresholds — and none of it has been pointed at this. Everything above is descriptive; there is no signal-loss number.
3. **Stage 7 self-suppresses on persistently contaminated channels**, because $\hat{\sigma}$ is fitted per channel. Stage 8 covers the steady case; a channel contaminated most-but-not-all of the time is covered by neither.
4. **The meteor statistic fires on non-propagation events.** It should require the FM/DTV rise to happen without a simultaneous transient elsewhere.
5. **Output is a union across antennas.** v0's product is per-input, and a box-air-only event currently masks box-gnd data too. Conservative, but it discards good data on a clean antenna — a decision to make deliberately before this is written in the `flags/<version>/` schema.
6. **The cross noise floor is a factor ~ 2 above radiometric and unexplained.**